

DDA3020 Machine Learning

Tutorial: Gaussian Mixture Models and the EM Algorithm

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Instructions

This exercise sheet covers the main ideas of the Gaussian Mixture Model (GMM) and the Expectation–Maximization (EM) algorithm, including: the formulation of a GMM as a linear superposition of Gaussian components; maximum-likelihood estimation for GMM and the MLE-based derivation of the update equations for the means $\boldsymbol{\mu}_k$, covariances $\boldsymbol{\Sigma}_k$, and mixing coefficients π_k ; the role of the responsibilities $\gamma_k^{(n)}$ as posterior probabilities / soft assignments; the comparison between GMM and K-means, including the $\boldsymbol{\Sigma}_k = \varepsilon \mathbf{I}$ limit that reduces EM for GMMs to K-means; the latent-variable view of GMM with $z \sim \text{Categorical}(\boldsymbol{\pi})$; Jensen’s inequality and its use in building a lower bound on the log-likelihood; the decomposition $\ln p(\mathcal{D}; \boldsymbol{\theta}) = \mathcal{L}(\mathbf{q}, \boldsymbol{\theta}) + \text{KL}(\mathbf{q} \| p(\mathbf{z} | \mathcal{D}; \boldsymbol{\theta}))$ and its use in deriving the general EM algorithm (E-step and M-step); and the monotonic-ascent convergence property of EM. You should be able to derive the E- and M-step updates from scratch, to use a Lagrange multiplier to enforce the constraint $\sum_k \pi_k = 1$, and to explain why the EM algorithm monotonically increases the log-likelihood at every iteration.

Useful Formulas

1. Gaussian mixture density.

$$p(\mathbf{x}) = \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k), \quad \pi_k \geq 0, \quad \sum_{k=1}^K \pi_k = 1.$$

2. Multivariate Gaussian.

$$\mathcal{N}(\mathbf{x} | \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{\sqrt{(2\pi)^d |\boldsymbol{\Sigma}|}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right).$$

3. Log-likelihood of a GMM.

$$\ln p(\mathcal{D} | \boldsymbol{\pi}, \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \sum_{n=1}^N \ln \left(\sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x}^{(n)} | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) \right).$$

4. Latent-variable formulation of GMM. With $z \in \{1, \dots, K\}$ denoting the latent component label,

$$p(z = k) = \pi_k, \quad p(\mathbf{x} | z = k) = \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k), \quad p(\mathbf{x}) = \sum_{k=1}^K p(z = k) p(\mathbf{x} | z = k).$$

5. Responsibilities (E-step for GMM).

$$\gamma_k^{(n)} \equiv p(z^{(n)} = k | \mathbf{x}^{(n)}; \boldsymbol{\theta}) = \frac{\pi_k \mathcal{N}(\mathbf{x}^{(n)} | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_{j=1}^K \pi_j \mathcal{N}(\mathbf{x}^{(n)} | \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)}.$$

6. Effective cluster size.

$$N_k = \sum_{n=1}^N \gamma_k^{(n)}, \quad \sum_{k=1}^K N_k = N.$$

7. M-step updates for GMM.

$$\boldsymbol{\mu}_k = \frac{1}{N_k} \sum_{n=1}^N \gamma_k^{(n)} \mathbf{x}^{(n)}, \quad \boldsymbol{\Sigma}_k = \frac{1}{N_k} \sum_{n=1}^N \gamma_k^{(n)} (\mathbf{x}^{(n)} - \boldsymbol{\mu}_k)(\mathbf{x}^{(n)} - \boldsymbol{\mu}_k)^\top, \quad \pi_k = \frac{N_k}{N}.$$

8. Jensen's inequality.

For a random variable X and a function f ,

$$f \text{ convex} \implies f(\mathbb{E}[X]) \leq \mathbb{E}[f(X)], \quad f \text{ concave} \implies f(\mathbb{E}[X]) \geq \mathbb{E}[f(X)].$$

9. ELBO / KL decomposition of the log-likelihood.

For any auxiliary distribution $\mathbf{q}(\mathbf{z})$,

$$\ln p(\mathcal{D}; \boldsymbol{\theta}) = \mathcal{L}(\mathbf{q}, \boldsymbol{\theta}) + \text{KL}(\mathbf{q}(\mathbf{z}) \parallel p(\mathbf{z} \mid \mathcal{D}; \boldsymbol{\theta})),$$

where

$$\mathcal{L}(\mathbf{q}, \boldsymbol{\theta}) = \sum_{\mathbf{z}} \mathbf{q}(\mathbf{z}) \ln \frac{p(\mathcal{D}, \mathbf{z}; \boldsymbol{\theta})}{\mathbf{q}(\mathbf{z})}, \quad \text{KL}(\mathbf{q} \parallel p) = - \sum_{\mathbf{z}} \mathbf{q}(\mathbf{z}) \ln \frac{p(\mathbf{z} \mid \mathcal{D}; \boldsymbol{\theta})}{\mathbf{q}(\mathbf{z})} \geq 0.$$

Consequently $\ln p(\mathcal{D}; \boldsymbol{\theta}) \geq \mathcal{L}(\mathbf{q}, \boldsymbol{\theta})$, with equality iff $\mathbf{q}(\mathbf{z}) = p(\mathbf{z} \mid \mathcal{D}; \boldsymbol{\theta})$.

10. General EM algorithm.

Given current parameters $\boldsymbol{\theta}^{\text{old}}$:

- **E-step:** set $q_n(z^{(n)}) = p(z^{(n)} \mid \mathbf{x}^{(n)}; \boldsymbol{\theta}^{\text{old}})$, which makes the bound tight: $\ln p(\mathcal{D}; \boldsymbol{\theta}^{\text{old}}) = \mathcal{L}(\mathbf{q}, \boldsymbol{\theta}^{\text{old}})$.
- **M-step:** update the parameters by maximizing the expected complete-data log-likelihood,

$$\boldsymbol{\theta}^{\text{new}} = \arg \max_{\boldsymbol{\theta}} \sum_{n=1}^N \mathbb{E}_{p(z^{(n)} \mid \mathbf{x}^{(n)}; \boldsymbol{\theta}^{\text{old}})} \left[\ln p(\mathbf{x}^{(n)}, z^{(n)}; \boldsymbol{\theta}) \right].$$

11. Monotonic ascent of EM.

$$\ln p(\mathcal{D}; \boldsymbol{\theta}^{\text{new}}) \geq \mathcal{L}(\mathbf{q}, \boldsymbol{\theta}^{\text{new}}) \geq \mathcal{L}(\mathbf{q}, \boldsymbol{\theta}^{\text{old}}) = \ln p(\mathcal{D}; \boldsymbol{\theta}^{\text{old}}).$$

12. Expected complete-data log-likelihood for GMM (objective maximized in the M-step).

$$Q(\boldsymbol{\theta}) = \sum_{n=1}^N \sum_{k=1}^K \gamma_k^{(n)} \left\{ \ln \pi_k + \ln \mathcal{N}(\mathbf{x}^{(n)} \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) \right\}.$$

Notation reminder. We write $\mathbf{x} \in \mathbb{R}^d$ for an observed data point and $z \in \{1, \dots, K\}$ for its latent component label. A Gaussian mixture model (GMM) is

$$p(\mathbf{x}) = \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k), \quad \pi_k \geq 0, \quad \sum_{k=1}^K \pi_k = 1.$$

Given a data set $\mathcal{D} = \{\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(N)}\}$ with corresponding latent variables $\mathbf{z} = \{z^{(1)}, \dots, z^{(N)}\}$, the *responsibility* of component k for observation n is

$$\gamma_k^{(n)} \equiv p(z^{(n)} = k \mid \mathbf{x}^{(n)}; \boldsymbol{\theta}) = \frac{\pi_k \mathcal{N}(\mathbf{x}^{(n)} \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_{j=1}^K \pi_j \mathcal{N}(\mathbf{x}^{(n)} \mid \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)},$$

and we set $N_k = \sum_{n=1}^N \gamma_k^{(n)}$. The complete-data log-likelihood is

$$\ln p(\mathcal{D}, \mathbf{z} \mid \boldsymbol{\mu}, \boldsymbol{\Sigma}, \boldsymbol{\pi}) = \sum_{n=1}^N \sum_{k=1}^K \mathbb{I}\{z^{(n)} = k\} \left\{ \ln \pi_k + \ln \mathcal{N}(\mathbf{x}^{(n)} \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) \right\}.$$

Additional Exercises

Q1. Consider a Gaussian mixture model expressed through a discrete latent variable $\mathbf{z}$ with 1-of- K encoding. Let the marginal distribution of $\mathbf{z}$ be

$$p(\mathbf{z}) = \prod_{k=1}^K \pi_k^{z_k}$$

and let the class-conditional distribution of $\mathbf{x}$ be

$$p(\mathbf{x} | \mathbf{z}) = \prod_{k=1}^K \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)^{z_k}.$$

Show that the marginal distribution $p(\mathbf{x}) = \sum_{\mathbf{z}} p(\mathbf{z}) p(\mathbf{x} | \mathbf{z})$ is a Gaussian mixture of the form

$$p(\mathbf{x}) = \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k).$$

Q2. Consider a special case of a Gaussian mixture model in which all component covariance matrices are constrained to have a common value, $\boldsymbol{\Sigma}_k = \boldsymbol{\Sigma}$ for every k . Using the latent-variable formulation, derive the M-step update equations that maximize the expected complete-data log-likelihood

$$\mathbb{E}_{\mathbf{z}}[\ln p(\mathcal{D}, \mathbf{z} | \boldsymbol{\mu}, \boldsymbol{\Sigma}, \boldsymbol{\pi})] = \sum_{n=1}^N \sum_{k=1}^K \gamma_k^{(n)} \{ \ln \pi_k + \ln \mathcal{N}(\mathbf{x}^{(n)} | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}) \}$$

with respect to the shared covariance $\boldsymbol{\Sigma}$. (Assume the responsibilities $\gamma_k^{(n)}$ and the component means $\boldsymbol{\mu}_k$ are held fixed.)

Q3. Consider the complete-data log-likelihood

$$\ln p(\mathcal{D}, \mathbf{z} | \boldsymbol{\mu}, \boldsymbol{\Sigma}, \boldsymbol{\pi}) = \sum_{n=1}^N \sum_{k=1}^K \mathbb{I}\{z^{(n)} = k\} \left\{ \ln \pi_k + \ln \mathcal{N}(\mathbf{x}^{(n)} | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) \right\},$$

in which every $z^{(n)} \in \{1, \dots, K\}$ is a *known* component label. Verify that maximizing this objective leads to the following results:

- (a) The means $\boldsymbol{\mu}_k$ and covariances $\boldsymbol{\Sigma}_k$ of each component are fitted independently to the corresponding group of data points (i.e. those n with $z^{(n)} = k$).
- (b) The mixing coefficients π_k are equal to the fraction of data points in each group, $\pi_k = N_k/N$, where N_k is the number of points assigned to component k , i.e. $N_k = \sum_{n=1}^N \mathbb{I}\{z^{(n)} = k\}$.

(Use a Lagrange multiplier to enforce $\sum_k \pi_k = 1$.)

Q4. Consider the expected complete-data log-likelihood of a GMM,

$$Q(\boldsymbol{\theta}) = \sum_{n=1}^N \sum_{k=1}^K \gamma_k^{(n)} \left\{ \ln \pi_k + \ln \mathcal{N}(\mathbf{x}^{(n)} | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) \right\},$$

where the responsibilities $\gamma_k^{(n)}$ are held fixed. Show that maximizing $Q(\boldsymbol{\theta})$ with respect to $\boldsymbol{\mu}_k$ yields the closed-form solution

$$\boldsymbol{\mu}_k = \frac{1}{N_k} \sum_{n=1}^N \gamma_k^{(n)} \mathbf{x}^{(n)}, \quad N_k = \sum_{n=1}^N \gamma_k^{(n)}.$$

Q5. With the same expected complete-data log-likelihood $Q(\boldsymbol{\theta})$ as in Exercise 4, show that maximizing $Q(\boldsymbol{\theta})$ with the responsibilities $\gamma_k^{(n)}$ held fixed yields:

(a) The covariance update

$$\boldsymbol{\Sigma}_k = \frac{1}{N_k} \sum_{n=1}^N \gamma_k^{(n)} (\mathbf{x}^{(n)} - \boldsymbol{\mu}_k)(\mathbf{x}^{(n)} - \boldsymbol{\mu}_k)^\top.$$

(b) The mixing-coefficient update

$$\pi_k = \frac{N_k}{N} \quad (\text{use a Lagrange multiplier for } \sum_k \pi_k = 1).$$

Q6. Consider a general mixture distribution

$$p(\mathbf{x}) = \sum_{k=1}^K \pi_k p(\mathbf{x} | k),$$

and partition $\mathbf{x}$ into two sub-vectors, $\mathbf{x} = (\mathbf{x}_a, \mathbf{x}_b)$. Show that the conditional density $p(\mathbf{x}_b | \mathbf{x}_a)$ is itself a mixture distribution of the form

$$p(\mathbf{x}_b | \mathbf{x}_a) = \sum_{k=1}^K \lambda_k p(\mathbf{x}_b | \mathbf{x}_a, k).$$

Give explicit expressions for the new mixing coefficients λ_k and the new component densities $p(\mathbf{x}_b | \mathbf{x}_a, k)$.

Q7. Consider a GMM in which every component has covariance $\boldsymbol{\Sigma}_k = \varepsilon \mathbf{I}$, where $\varepsilon > 0$ is a shared scalar. As $\varepsilon \rightarrow 0$, the responsibility $\gamma_k^{(n)}$ tends to an indicator $r_{nk} \in \{0, 1\}$ with $r_{nk} = 1$ iff component k is closest to $\mathbf{x}^{(n)}$ in Euclidean distance. Show that, in this limit, maximizing the expected complete-data log-likelihood

$$\sum_{n=1}^N \sum_{k=1}^K \gamma_k^{(n)} \{ \ln \pi_k + \ln \mathcal{N}(\mathbf{x}^{(n)} | \boldsymbol{\mu}_k, \varepsilon \mathbf{I}) \}$$

with respect to $\{\boldsymbol{\mu}_k\}$ is equivalent to minimizing the K-means distortion

$$J = \sum_{n=1}^N \sum_{k=1}^K r_{nk} \|\mathbf{x}^{(n)} - \boldsymbol{\mu}_k\|^2.$$

Q8. Let

$$p(\mathbf{x}) = \sum_{k=1}^K \pi_k p(\mathbf{x} | k)$$

be a mixture distribution in which each component $p(\mathbf{x} | k)$ has mean $\boldsymbol{\mu}_k$ and covariance $\boldsymbol{\Sigma}_k$. Show that the mean and covariance of the overall mixture are

$$\mathbb{E}[\mathbf{x}] = \sum_{k=1}^K \pi_k \boldsymbol{\mu}_k, \quad \text{cov}[\mathbf{x}] = \sum_{k=1}^K \pi_k \left\{ \boldsymbol{\Sigma}_k + \boldsymbol{\mu}_k \boldsymbol{\mu}_k^\top \right\} - \mathbb{E}[\mathbf{x}] \mathbb{E}[\mathbf{x}]^\top.$$

Q9. Let $\mathcal{D}$ be observed data and $\mathbf{z}$ denote the set of latent variables. For any distribution $\mathbf{q}(\mathbf{z})$ over the latent variables and any parameter value $\boldsymbol{\theta}$, define

$$\mathcal{L}(\mathbf{q}, \boldsymbol{\theta}) = \sum_{\mathbf{z}} \mathbf{q}(\mathbf{z}) \ln \frac{p(\mathcal{D}, \mathbf{z} | \boldsymbol{\theta})}{\mathbf{q}(\mathbf{z})}, \quad \text{KL}(\mathbf{q} \| p) = - \sum_{\mathbf{z}} \mathbf{q}(\mathbf{z}) \ln \frac{p(\mathbf{z} | \mathcal{D}, \boldsymbol{\theta})}{\mathbf{q}(\mathbf{z})}.$$

Verify the identity

$$\ln p(\mathcal{D} | \boldsymbol{\theta}) = \mathcal{L}(\mathbf{q}, \boldsymbol{\theta}) + \text{KL}(\mathbf{q}(\mathbf{z}) \| p(\mathbf{z} | \mathcal{D}, \boldsymbol{\theta})).$$

Q10. Let $\mathcal{L}(\mathbf{q}, \boldsymbol{\theta})$ and $\text{KL}(\mathbf{q} \| p(\mathbf{z} \mid \mathcal{D}, \boldsymbol{\theta}))$ be as in Exercise 9, so that

$$\ln p(\mathcal{D} \mid \boldsymbol{\theta}) = \mathcal{L}(\mathbf{q}, \boldsymbol{\theta}) + \text{KL}(\mathbf{q}(\mathbf{z}) \| p(\mathbf{z} \mid \mathcal{D}, \boldsymbol{\theta})). \quad (\star)$$

Suppose we set $\mathbf{q}(\mathbf{z}) = p(\mathbf{z} \mid \mathcal{D}, \boldsymbol{\theta}^{\text{old}})$ (the E-step choice). Show that, evaluated at $\boldsymbol{\theta} = \boldsymbol{\theta}^{\text{old}}$, the gradient of $\mathcal{L}(\mathbf{q}, \boldsymbol{\theta})$ with respect to $\boldsymbol{\theta}$ equals the gradient of $\ln p(\mathcal{D} \mid \boldsymbol{\theta})$.

Solutions

Q1. Answer: Using the 1-of- K representation of $\mathbf{z}$,

$$p(\mathbf{x}) = \sum_{\mathbf{z}} p(\mathbf{x} | \mathbf{z}) p(\mathbf{z}) = \sum_{\mathbf{z}} \prod_{k=1}^K (\pi_k \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k))^{z_k}.$$

Because $\mathbf{z}$ is one-hot, the K possible states of $\mathbf{z}$ can be indexed by an integer $j \in \{1, \dots, K\}$ (the index of the single nonzero coordinate). Writing $\mathbf{I}_{kj} = 1$ if $k = j$ and 0 otherwise, we obtain

$$p(\mathbf{x}) = \sum_{j=1}^K \prod_{k=1}^K (\pi_k \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k))^{\mathbf{I}_{kj}} = \sum_{j=1}^K \pi_j \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j),$$

which is the Gaussian mixture of the desired form. $\square$

Q2. Answer: In this case, the expected complete-data log likelihood function becomes

$$\mathbb{E}_{\mathbf{z}}[\ln p(\mathcal{D}, \mathbf{z} | \boldsymbol{\mu}, \boldsymbol{\Sigma}, \boldsymbol{\pi})] = \sum_{n=1}^N \sum_{k=1}^K \gamma_k^{(n)} \{\ln \pi_k + \ln \mathcal{N}(\mathbf{x}^{(n)} | \boldsymbol{\mu}_k, \boldsymbol{\Sigma})\}.$$

Differentiating this with respect to $\boldsymbol{\Sigma}^{-1}$ (using the standard matrix-calculus identities $\partial \ln |\boldsymbol{\Sigma}^{-1}| / \partial \boldsymbol{\Sigma}^{-1} = \boldsymbol{\Sigma}$ and $\partial \text{tr}(\boldsymbol{\Sigma}^{-1} \mathbf{A}) / \partial \boldsymbol{\Sigma}^{-1} = \mathbf{A}$), we get

$$\frac{N}{2} \boldsymbol{\Sigma} - \frac{1}{2} \sum_{n=1}^N \sum_{k=1}^K \gamma_k^{(n)} (\mathbf{x}^{(n)} - \boldsymbol{\mu}_k)(\mathbf{x}^{(n)} - \boldsymbol{\mu}_k)^\top,$$

where we have used that $\sum_{k=1}^K \gamma_k^{(n)} = 1$ for all n . Setting this equal to zero and rearranging, we obtain

$$\boldsymbol{\Sigma} = \frac{1}{N} \sum_{n=1}^N \sum_{k=1}^K \gamma_k^{(n)} (\mathbf{x}^{(n)} - \boldsymbol{\mu}_k)(\mathbf{x}^{(n)} - \boldsymbol{\mu}_k)^\top. \quad \square$$

Q3. Answer: Consider first the optimization with respect to the parameters $\{\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k\}$. For this, we can ignore the terms which depend on $\ln \pi_k$. We note that, for each data point n , the known label $z^{(n)}$ selects exactly one component. Let

$$\mathcal{I}_k = \{n : z^{(n)} = k\}$$

be the index set of data points assigned to component k . The complete-data log-likelihood function can then be written

$$\ln p(\mathcal{D}, \mathbf{z} | \boldsymbol{\mu}, \boldsymbol{\Sigma}, \boldsymbol{\pi}) = \sum_{k=1}^K \left\{ \sum_{n \in \mathcal{I}_k} \ln \mathcal{N}(\mathbf{x}^{(n)} | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) \right\}.$$

This represents the sum of K independent terms, one for each component in the mixture. When we maximize this term with respect to $\boldsymbol{\mu}_k$ and $\boldsymbol{\Sigma}_k$ we will simply be fitting the k -th component to the data points indexed by $\mathcal{I}_k$, for which we will obtain the usual maximum-likelihood results for a single Gaussian.

For the mixing coefficients we need only consider the terms in $\ln \pi_k$, but we must introduce a Lagrange multiplier to handle the constraint $\sum_k \pi_k = 1$. Thus we maximize

$$\sum_{n=1}^N \sum_{k=1}^K \mathbb{I}\{z^{(n)} = k\} \ln \pi_k + \lambda \left(\sum_{k=1}^K \pi_k - 1 \right),$$

which gives

$$0 = \sum_{n=1}^N \frac{\mathbb{I}\{z^{(n)} = k\}}{\pi_k} + \lambda.$$

Multiplying through by π_k and summing over k we obtain $\lambda = -N$, from which we have

$$\pi_k = \frac{1}{N} \sum_{n=1}^N \mathbb{I}\{z^{(n)} = k\} = \frac{N_k}{N},$$

where N_k is the number of data points with $z^{(n)} = k$. $\square$

Q 4. Answer: Using the quadratic form of the Gaussian log-density, we can write the $\boldsymbol{\mu}$ -dependent part of $Q(\boldsymbol{\theta})$ as

$$-\frac{1}{2} \sum_{n=1}^N \sum_{j=1}^K \gamma_j^{(n)} (\mathbf{x}^{(n)} - \boldsymbol{\mu}_j)^\top \boldsymbol{\Sigma}_j^{-1} (\mathbf{x}^{(n)} - \boldsymbol{\mu}_j) + \text{const.},$$

where “const.” summarizes terms independent of $\boldsymbol{\mu}_j$ (for all j). Taking the derivative of this with respect to $\boldsymbol{\mu}_k$, we get

$$-\sum_{n=1}^N \gamma_k^{(n)} (\boldsymbol{\Sigma}_k^{-1} \boldsymbol{\mu}_k - \boldsymbol{\Sigma}_k^{-1} \mathbf{x}^{(n)}).$$

Setting this to zero and rearranging yields

$$\boldsymbol{\mu}_k = \frac{1}{N_k} \sum_{n=1}^N \gamma_k^{(n)} \mathbf{x}^{(n)}, \quad N_k = \sum_{n=1}^N \gamma_k^{(n)}. \quad \square$$

Q 5. Answer: *Covariance.* If we differentiate $Q(\boldsymbol{\theta})$ with respect to $\boldsymbol{\Sigma}_k^{-1}$, while keeping the responsibilities $\gamma_k^{(n)}$ fixed, we get

$$\frac{\partial}{\partial \boldsymbol{\Sigma}_k^{-1}} Q(\boldsymbol{\theta}) = \sum_{n=1}^N \gamma_k^{(n)} \frac{1}{2} \left\{ \boldsymbol{\Sigma}_k - (\mathbf{x}^{(n)} - \boldsymbol{\mu}_k)(\mathbf{x}^{(n)} - \boldsymbol{\mu}_k)^\top \right\},$$

where we have used the standard matrix-derivative identities. Setting this equal to zero and rearranging gives

$$\boldsymbol{\Sigma}_k = \frac{1}{N_k} \sum_{n=1}^N \gamma_k^{(n)} (\mathbf{x}^{(n)} - \boldsymbol{\mu}_k)(\mathbf{x}^{(n)} - \boldsymbol{\mu}_k)^\top.$$

Mixing coefficients. For π_k , we add a Lagrange multiplier term to $Q(\boldsymbol{\theta})$ to enforce the constraint $\sum_{k=1}^K \pi_k = 1$, yielding

$$Q(\boldsymbol{\theta}) + \lambda \left(\sum_{k=1}^K \pi_k - 1 \right).$$

Differentiating this with respect to π_k , we get

$$\sum_{n=1}^N \gamma_k^{(n)} \frac{1}{\pi_k} + \lambda = \frac{N_k}{\pi_k} + \lambda,$$

where we have used the definition of N_k . Setting this equal to zero and rearranging gives $N_k = -\pi_k \lambda$. Summing both sides over k and using $\sum_k \pi_k = 1$ and $\sum_k N_k = N$, we see that $-\lambda = N$ and hence

$$\pi_k = \frac{N_k}{N}. \quad \square$$

Q6. Answer: For the mixture model the joint distribution of $(\mathbf{x}_a, \mathbf{x}_b)$ can be written

$$p(\mathbf{x}_a, \mathbf{x}_b) = \sum_{k=1}^K \pi_k p(\mathbf{x}_a, \mathbf{x}_b | k).$$

We can find the conditional density $p(\mathbf{x}_b | \mathbf{x}_a)$ by making use of the relation

$$p(\mathbf{x}_b | \mathbf{x}_a) = \frac{p(\mathbf{x}_a, \mathbf{x}_b)}{p(\mathbf{x}_a)}.$$

For the mixture model the marginal density of $\mathbf{x}_a$ is given by

$$p(\mathbf{x}_a) = \sum_{k=1}^K \pi_k p(\mathbf{x}_a | k), \quad p(\mathbf{x}_a | k) = \int p(\mathbf{x}_a, \mathbf{x}_b | k) d\mathbf{x}_b.$$

Thus we can write the conditional density in the form

$$p(\mathbf{x}_b | \mathbf{x}_a) = \frac{\sum_{k=1}^K \pi_k p(\mathbf{x}_a, \mathbf{x}_b | k)}{\sum_{j=1}^K \pi_j p(\mathbf{x}_a | j)}.$$

Now we decompose the numerator using $p(\mathbf{x}_a, \mathbf{x}_b | k) = p(\mathbf{x}_b | \mathbf{x}_a, k) p(\mathbf{x}_a | k)$, which allows us finally to write the conditional density as a mixture model of the form

$$p(\mathbf{x}_b | \mathbf{x}_a) = \sum_{k=1}^K \lambda_k p(\mathbf{x}_b | \mathbf{x}_a, k),$$

where the mixture coefficients are given by

$$\lambda_k \equiv p(k | \mathbf{x}_a) = \frac{\pi_k p(\mathbf{x}_a | k)}{\sum_{j=1}^K \pi_j p(\mathbf{x}_a | j)},$$

and $p(\mathbf{x}_b | \mathbf{x}_a, k)$ is the conditional for component k . □

Q7. Answer: As discussed in the lecture, $\gamma_k^{(n)} \rightarrow r_{nk}$ as $\varepsilon \rightarrow 0$ (the soft assignments become hard). Since $\Sigma_k = \varepsilon \mathbf{I}$ for all k are no longer free parameters, and π_k equals the proportion of data points assigned to cluster k (which, under a reasonable initialization of $\boldsymbol{\pi}$ and $\{\boldsymbol{\mu}_k\}$, remains strictly positive), we can maximize the expected complete-data log likelihood with respect to $\{\boldsymbol{\mu}_k\}$ independently of $\boldsymbol{\pi}$. What remains is

$$\sum_{n=1}^N \sum_{k=1}^K r_{nk} \ln \mathcal{N}(\mathbf{x}^{(n)} | \boldsymbol{\mu}_k, \varepsilon \mathbf{I}) = \sum_{n=1}^N \sum_{k=1}^K r_{nk} \left(-\frac{1}{2\varepsilon} \|\mathbf{x}^{(n)} - \boldsymbol{\mu}_k\|^2 \right) + \text{const.},$$

which equals the negative of J up to a positive scaling factor (which is independent of $\{\boldsymbol{\mu}_k\}$). Hence maximizing the expected complete-data log-likelihood in this limit is equivalent to minimizing J . □

Q8. Answer: Since the expectation of a sum is the sum of the expectations, we have

$$\mathbb{E}[\mathbf{x}] = \sum_{k=1}^K \pi_k \mathbb{E}_k[\mathbf{x}] = \sum_{k=1}^K \pi_k \boldsymbol{\mu}_k,$$

where $\mathbb{E}_k[\cdot]$ denotes the expectation of $\mathbf{x}$ under the distribution $p(\mathbf{x} | k)$. To find the covariance we use the general relation

$$\text{cov}[\mathbf{x}] = \mathbb{E}[\mathbf{x}\mathbf{x}^\top] - \mathbb{E}[\mathbf{x}] \mathbb{E}[\mathbf{x}]^\top$$

to give

$$\text{cov}[\mathbf{x}] = \sum_{k=1}^K \pi_k \mathbb{E}_k[\mathbf{x}\mathbf{x}^\top] - \mathbb{E}[\mathbf{x}] \mathbb{E}[\mathbf{x}]^\top = \sum_{k=1}^K \pi_k \left\{ \boldsymbol{\Sigma}_k + \boldsymbol{\mu}_k \boldsymbol{\mu}_k^\top \right\} - \mathbb{E}[\mathbf{x}] \mathbb{E}[\mathbf{x}]^\top. \quad \square$$

Q9. Answer: This is analogous to Solution 10.1, with the integrals replaced by sums. Starting from the definition of $\mathcal{L}(\mathbf{q}, \boldsymbol{\theta})$, we use the product rule $p(\mathcal{D}, \mathbf{z} | \boldsymbol{\theta}) = p(\mathbf{z} | \mathcal{D}, \boldsymbol{\theta}) p(\mathcal{D} | \boldsymbol{\theta})$ together with the definition of $\text{KL}(\mathbf{q} \| p)$ to get

$$\begin{aligned} \mathcal{L}(\mathbf{q}, \boldsymbol{\theta}) &= \sum_{\mathbf{z}} \mathbf{q}(\mathbf{z}) \ln \left\{ \frac{p(\mathcal{D}, \mathbf{z} | \boldsymbol{\theta})}{\mathbf{q}(\mathbf{z})} \right\} \\ &= \sum_{\mathbf{z}} \mathbf{q}(\mathbf{z}) \ln \left\{ \frac{p(\mathbf{z} | \mathcal{D}, \boldsymbol{\theta}) p(\mathcal{D} | \boldsymbol{\theta})}{\mathbf{q}(\mathbf{z})} \right\} \\ &= \sum_{\mathbf{z}} \mathbf{q}(\mathbf{z}) \left(\ln \left\{ \frac{p(\mathbf{z} | \mathcal{D}, \boldsymbol{\theta})}{\mathbf{q}(\mathbf{z})} \right\} + \ln p(\mathcal{D} | \boldsymbol{\theta}) \right) \\ &= -\text{KL}(\mathbf{q} \| p(\mathbf{z} | \mathcal{D}, \boldsymbol{\theta})) + \ln p(\mathcal{D} | \boldsymbol{\theta}). \end{aligned}$$

Rearranging this, we immediately get the decomposition $\ln p(\mathcal{D} | \boldsymbol{\theta}) = \mathcal{L}(\mathbf{q}, \boldsymbol{\theta}) + \text{KL}(\mathbf{q} \| p(\mathbf{z} | \mathcal{D}, \boldsymbol{\theta}))$. Because the KL divergence is always non-negative, we also obtain $\ln p(\mathcal{D} | \boldsymbol{\theta}) \geq \mathcal{L}(\mathbf{q}, \boldsymbol{\theta})$, with equality if and only if $\mathbf{q}(\mathbf{z}) = p(\mathbf{z} | \mathcal{D}, \boldsymbol{\theta})$. $\square$

Q10. Answer: This follows from the fact that the Kullback–Leibler divergence, $\text{KL}(\mathbf{q} \| p)$, is at its minimum ($= 0$) when $\mathbf{q}$ and p are identical. By the E-step choice, $\mathbf{q}(\mathbf{z}) = p(\mathbf{z} | \mathcal{D}, \boldsymbol{\theta}^{\text{old}})$, so

$$\text{KL}(\mathbf{q}(\mathbf{z}) \| p(\mathbf{z} | \mathcal{D}, \boldsymbol{\theta}))$$

is a non-negative function of $\boldsymbol{\theta}$ that attains its minimum value 0 at $\boldsymbol{\theta} = \boldsymbol{\theta}^{\text{old}}$. Therefore

$$\left. \frac{\partial}{\partial \boldsymbol{\theta}} \text{KL}(\mathbf{q} \| p(\mathbf{z} | \mathcal{D}, \boldsymbol{\theta})) \right|_{\boldsymbol{\theta} = \boldsymbol{\theta}^{\text{old}}} = 0,$$

since $p(\mathbf{z} | \mathcal{D}, \boldsymbol{\theta})$ depends on $\boldsymbol{\theta}$. Therefore, if we compute the gradient of both sides of the identity $(\star)$ in Exercise 9 with respect to $\boldsymbol{\theta}$, the contribution from the KL term on the right-hand side, evaluated at $\boldsymbol{\theta}^{\text{old}}$, is zero, and so the gradient of $\mathcal{L}(\mathbf{q}, \boldsymbol{\theta})$ must equal that of $\ln p(\mathcal{D} | \boldsymbol{\theta})$ at that point. $\square$